

INC-2026-0312 - Duplicate Reversal Entries in Settlement Ledger

Incident Report | Engineering

Document ID	INC-2026-0312	Seller ID	S001
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-03-16
Related Scenario ID	SC-001	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0312
Severity	SEV-2
Service	Settlement and Payout Service
Component	settlement_ledger write path
Detected	2026-03-12 10:35 IST
Mitigated	2026-03-13 15:45 IST
Resolved	2026-03-14 16:30 IST
Incident owner	Vikram Singh, Engineering
Seller affected	S001 - Aurelia Home Decor
Related ticket	TKT-2026-0312-114

The settlement batch covering 5 to 9 March 2026 paid seller S001 INR 4,82,000 less than the correct amount. 1,914 return reversal entries were written to the settlement ledger twice, so each return was deducted from the seller's payout twice.

The duplication was introduced by a deduplication key that no longer identified reversal entries uniquely, and was triggered when the batch was reprocessed following a transient job failure.

2. Impact

Dimension	Detail
Financial	INR 4,82,000 under-payment to a single seller
Ledger entries affected	3,828 written where 1,914 were correct
Sellers affected	1 (S001)
Payout delay	2 days beyond the scheduled payout date
Customer impact	None
Data loss	None; all entries were recoverable and reversible

Only S001 was affected. The trigger required a reprocessing run, and the 5 to 9 March batch for this seller was the only batch reprocessed in the affected period. Other sellers processed on the same code path but

were never reprocessed, so the latent defect did not surface for them.

That distinction matters: the defect was platform-wide, the impact was seller-specific, and the difference between the two was a transient job failure that could have occurred on any account.

3. Detection

The incident was detected by the seller, not by the platform.

Aurelia’s finance team reconciles every settlement cycle against its own order export and flags variances above one percent. They identified the shortfall on the morning of 12 March and raised TKT-2026-0312-114 at 09:48 IST with a reconciliation spreadsheet attached that had already isolated the reversal block as the source of the difference.

Support confirmed duplicate ledger entries at 10:35 IST and escalated to Engineering at 11:20 IST.

Time from occurrence to detection: approximately 19 hours. There was no automated control that would have detected this. Settlement variance was not monitored.

4. Timeline

Time (IST)	Event
11 Mar 15:42	Settlement batch job for 5-9 Mar starts
11 Mar 15:58	Job fails with a transient database connection timeout
11 Mar 16:05	Automatic retry triggers reprocessing of the same batch
11 Mar 16:31	Reprocessing completes; duplicate reversal entries written
11 Mar 23:00	Payout computed from the incorrect ledger position
12 Mar 09:48	Seller raises ticket with reconciliation evidence
12 Mar 10:35	Support confirms 1,914 reversals appearing twice
12 Mar 11:20	Escalated to Engineering; incident opened
12 Mar 12:15	Reprocessing run identified as the trigger
12 Mar 14:10	Deduplication key defect confirmed in v1.9 schema
12 Mar 16:00	Payout for the cycle held pending correction
13 Mar 11:20	Fix developed and reviewed
13 Mar 15:45	Hotfix deployed; affected batch reprocessed under a replay token
14 Mar 09:30	Operations verification against Payment Recovery Runbook (OPS-RB-004)
14 Mar 12:20	Corrected payout figure confirmed
14 Mar 16:30	INR 4,82,000 credited to seller

5. Symptoms

- Settlement report for the cycle showed 3,828 reversal entries against 1,914 returns in the order history.
- Each duplicate pair referenced the same order and the same return, but carried distinct ledger entry identifiers.
- Gross merchandise value, commission and adjustment blocks reconciled

exactly. Only the reversal block was wrong.

- The batch job log reported success. There was no error to investigate.

The clean reconciliation of every other block was the most useful diagnostic signal. It ruled out a calculation problem and pointed directly at entry creation.

6. Affected Services

Service	Role in the incident	State
Settlement and Payout Service	Wrote the duplicate entries	Defective
settlement_ledger store	Accepted the duplicates	Functioning as designed
Batch scheduler	Triggered the retry	Functioning as designed
Payout execution	Paid the incorrect computed amount	Functioning as designed
Order and returns services	Source of truth for returns	Correct throughout

No service was unavailable at any point. This was a correctness failure, not an availability failure, which is why no monitoring alert fired.

7. Investigation

The investigation followed three questions.

Was the returns data correct? Yes. The order and returns services showed 1,914 returns for the period, matching the seller's own export exactly. The input was never in question.

Why were there two entries per return? Each pair had distinct entry identifiers and identical business content. Creation timestamps clustered in two groups, at 15:51 and at 16:22 on 11 March, which mapped to the original run and the retry.

Why did the retry not deduplicate? The ledger writes entries through an idempotent upsert keyed on a composite deduplication key. Inspecting the key construction in the v1.9 schema showed:

```
dedupe_key = hash(settlement_cycle_id, seller_id, entry_type)
```

For a sale entry this is sufficient, because a cycle produces one aggregate sale entry per seller. For a reversal entry it is not: a cycle can contain many reversals for one seller, all sharing cycle, seller and entry type. The key therefore could not distinguish one reversal from another.

The v1.8 key had been:

```
dedupe_key = hash(settlement_cycle_id, seller_id, entry_type, order_id)
```

The order_id component was dropped during the v1.8 to v1.9 migration. The migration changed the key to accommodate a new aggregate entry type that has no order identifier. The author made the field optional in the schema and removed it from the key rather than making the key type-dependent.

Because the upsert matched nothing on the retry, it inserted rather than updated.

8. Root Cause

The deduplication key for settlement ledger entries omitted order_id after the v1.8 to v1.9 schema migration. For reversal entries, which are order-scoped, this made the key non-unique. When the 5-9 March batch was reprocessed after a transient failure, every reversal entry was inserted a second time instead of being recognised as already present, double-deducting 1,914 returns and reducing the payout by INR 4,82,000.

9. Contributing Factors

1. **Reprocessing was not idempotent in practice.** The design assumed idempotency from the deduplication key alone, with no independent safeguard against a batch being processed twice.
2. **The migration was not tested against a reprocessing path.** Test coverage exercised first-run behaviour for all entry types. No test reprocessed a batch containing reversals.
3. **No settlement variance monitoring existed.** A payout differing from order and return history by 4.5 percent produced no alert.
4. **A schema change altered financial correctness without financial review.**
The migration was reviewed as a schema change. Its effect on ledger uniqueness guarantees was not examined.
5. **A successful job log masked an incorrect outcome.** Both runs reported success, because both did what they were asked to do.

10. Mitigation

Immediate mitigation was to hold the payout for the cycle at 16:00 on 12 March, once duplication was confirmed but before the corrected figure was known. This prevented the incorrect amount from being treated as final while the investigation continued.

No traffic shifting or rollback was applicable. The defect was in data already written, not in a running request path.

11. Resolution

The deduplication key was restored to include the order identifier, with a type-dependent construction so that aggregate entry types without an order identifier remain valid:

- For order-scoped entry types: ``hash(cycle_id, seller_id, entry_type, order_id)``.
- For aggregate entry types: `hash(cycle_id, seller_id, entry_type)`.

The duplicate entries for the affected batch were removed and the batch was reprocessed under an explicit replay token. The corrected payout was verified by Operations against the Payment Recovery Runbook (OPS-RB-004), and the shortfall of INR 4,82,000 was credited on 14 March 2026. The controlling rule is the uniqueness requirement for ledger entries under the Payment and Settlement Policy (POL-FIN-002).

12. Validation

Check	Method	Result
Duplicate entries removed	Count of ledger entries per return for the cycle	1 per return, 1,914 total
Corrected payout accurate	Reconciled against seller's own order export	Matched to the rupee
Fix prevents recurrence	Batch deliberately reprocessed three times in staging	No duplicates on runs 2 and 3
Aggregate entries unaffected	Aggregate entry types processed in staging	Correct, no regression

Check	Method	Result
Historical exposure	Twelve months of batches back-tested for duplicates	Two low-value anomalies from 2025, both corrected

The back-test was the most important validation step. It established that the defect had produced a small number of earlier errors that had never been detected, which changed the follow-up from a single fix into a controls project.

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Type-dependent deduplication key	Vikram Singh	2026-03-13	Complete
PA-2	Ledger invariant checks before payout authorisation	Vikram Singh	2026-05-29	Complete
PA-3	Automated settlement variance detection	Vikram Singh	2026-07-10	Complete
PA-4	Explicit replay token for batch reprocessing	Neha Gupta	2026-07-31	Complete
PA-5	Financial review required for ledger schema changes	Engineering	2026-04-30	Complete
PA-6	Reprocessing test cases for every entry type	Vikram Singh	2026-04-17	Complete

13.1 On PA-2 and PA-4

PA-2 introduces three invariants checked before any payout is authorised: every reversal entry maps to exactly one originating order line, entry types sum to the batch net, and no entry appears more than once under its composite key. The third invariant is the one that would have caught this incident.

PA-4 removes the trigger condition entirely. A batch now carries a processing token and a second run with the same token is rejected unless an operator explicitly authorises a replay with a documented reason. Correctness no longer depends solely on the deduplication key being right.

14. Follow-Up

The broader finding from this incident was not the key defect. It was that the platform had no way to know a settlement was wrong until a seller reconciled it manually. That finding opened the settlement reconciliation hardening project (PRJ-S001-SRH), which tracks PA-2, PA-3 and PA-4 along with a seller-facing variance statement.

Post-incident review held 16 March 2026 with Engineering, Operations, Support and Business. The review accepted the root cause without dispute and concentrated on detection rather than on the defect itself.